

JOONWON LEE

Piscataway, NJ 08854 | 612-438-9144 | joonwon.lee22@gmail.com | github.com/cl20813

SUMMARY

Statistics Ph.D. candidate and quantitative researcher specializing in stochastic modeling, likelihood-based inference, Monte Carlo simulation, hypothesis testing, and independent model validation. Develops scalable Python/PyTorch analytical systems for complex dependent data and communicates model assumptions, performance, and risk tradeoffs clearly. Brings financial model-risk and insurance-pricing experience with a strong interest in capital-markets pricing, portfolio risk, and quantitative research.

EDUCATION

Ph.D. in Statistics, Rutgers University, Piscataway, NJ

Expected May 2027 | GPA: 3.8/4.0

M.S. in Statistics, University of Minnesota, Minneapolis, MN

Sep 2021 | GPA: 3.7/4.0

TECHNICAL SKILLS

Programming and Platforms: Python (NumPy, Pandas, SciPy, PyTorch, LightGBM, scikit-learn), C++/pybind11 integration, SQL, R, Linux, Git, AWS EC2, HPC/SLURM; CPU/GPU computing, cached artifacts, and restartable analytical pipelines

Quantitative Methods: Probability and stochastic processes, Monte Carlo simulation, hypothesis testing, MLE/GLS, numerical optimization, linear algebra, regression, GLM/GBM, machine learning, time-series and spatiotemporal modeling, model selection, validation, and performance diagnostics

QUANTITATIVE INDUSTRY EXPERIENCE

JPMorgan Chase - Summer Quantitative Analytics Associate, Model Risk Governance and Review

Summer 2026

- Developed and independently validated a Sequential Probability Ratio Test for binary risk indicators, translating a monitoring problem into explicit hypotheses, likelihood-ratio boundaries, and controlled decisions.
- Combined Brownian-motion approximation, dynamic-programming probability propagation, and simulation to evaluate boundary-crossing probabilities, stopping behavior, Type I/II error, and early-decision tradeoffs; documented model assumptions, performance, and limitations.

Travelers - Data Science Leadership Program, Business Insurance Pricing Team

Summer 2025

- Developed an end-to-end LightGBM pricing pipeline across 2.46M+ records, including data preparation, feature processing, GLM benchmarking, model selection, validation, and performance diagnostics on AWS EC2.
- Translated model diagnostics and feature-attribution findings into pricing and risk-segmentation recommendations for business stakeholders.

QUANTITATIVE RESEARCH AND MARKETS

Ph.D. Dissertation Research - Scalable Stochastic Modeling and Model Diagnostics

Sep 2024 - Present

- **Scalable Inference:** Developed an advection-aware Vecchia likelihood approximation for nonseparable Gaussian processes, replacing dense covariance operations with ordered conditional models to fit up to 145,008 observations per day.
- **Model Diagnostics:** Designed scale- and frequency-resolved tools to identify whether discrepancies arise from low- or high-frequency behavior, missingness, measurement artifacts, or covariance-model misspecification.
- **Research Engineering:** Built reusable Python/PyTorch workflows for data-quality filtering, optimization, simulation, cached outputs, and restartable CPU/GPU and HPC execution; integrated performance-critical C++ ordering routines through pybind11.

IMC Prosperity Algorithmic Trading Competition - Top 2.39% of Participants

Apr 2025

- Built and evaluated a market-making strategy using market-based fair-value estimation, inventory-aware quoting, rolling-window exposure controls, and dynamic liquidation rules to manage position risk.

ADDITIONAL

Coursework: Machine Learning, Linear Algebra, Probability Theory, Stochastic Processes, Statistical Computing, Advanced Theory of Statistics I-II, Data Structures and Algorithms, Microeconomics, Econometrics

Languages: Korean (Native); English (Fluent); Chinese (Basic)